

ASWAN INTL, Egypt

WMO: 624140

Lat: 23.964N

Lon: 32.820E

Elev: 662

StdP: 14.35

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.6	48.7	15.5	12.3	72.4	19.0	14.7	73.5	20.4	67.4	18.5	66.0	6.6	10	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	111.7	69.7	109.8	69.3	108.0	68.7	72.7	105.2	71.4	104.7	70.4	103.8	10.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	82.1	88.6	58.0	73.6	91.0	55.7	67.6	91.8	36.7	105.2	35.5	104.5	34.6	104.0	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	17.8	16.1	41.5	115.3	3.4	1.6	39.0	116.5	37.1	117.4	35.2	118.3	32.7	119.5
			34.8	76.2	1.8	1.8	33.5	77.6	32.4	78.6	31.4	79.7	30.1	81.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	62.5	66.2	73.6	82.3	89.8	93.7	95.0	94.9	91.3	84.7	73.6	64.7	(6)
(7)		DBStd	12.95	5.62	6.45	6.68	6.60	5.39	3.72	3.02	2.80	3.36	5.67	5.78	5.42	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	258	120	54	7	0	0	0	0	0	0	0	5	72	(9)
(10)		CDD50	11348	387	455	730	969	1233	1310	1394	1393	1238	1076	708	455	(10)
(11)		CDD65	6131	42	89	272	520	768	860	929	928	788	611	263	61	(11)
(12)		CDH74	93881	438	1050	3213	7181	12132	14455	15968	15737	12387	8168	2590	562	(12)
(13)		CDH80	61927	109	418	1545	4322	8155	10222	11520	11286	8263	4849	1083	155	(13)
(14)	Wind	WSAvg	8.7	7.9	8.3	8.6	8.9	9.1	9.5	9.4	9.5	8.9	8.5	8.1	7.9	(14)
(15)	Precipitation	PrecAvg	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(15)
(16)		PrecMax	0.7	0.1	0.0	0.6	0.4	0.1	0.0	0.0	0.0	0.1	0.4	0.0	0.1	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.2	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	87.6	95.3	102.1	109.1	113.6	114.7	114.1	113.1	111.0	107.9	98.3	89.3	(19)	
(20)		MCWB	63.1	63.7	66.5	67.7	68.6	69.0	71.2	71.0	70.5	68.6	66.9	65.1	(20)	
(21)		DB	82.7	89.8	97.0	105.4	109.8	111.5	111.1	110.5	107.8	104.3	93.5	84.3	(21)	
(22)		MCWB	61.6	61.4	63.8	66.9	68.6	68.8	69.9	70.5	69.5	68.7	65.9	62.6	(22)	
(23)		DB	79.1	85.6	93.0	101.9	107.2	109.3	109.1	108.2	105.7	100.9	89.7	80.2	(23)	
(24)		MCWB	58.8	59.3	61.8	64.9	67.7	68.2	69.2	69.6	68.8	68.0	64.3	60.5	(24)	
(25)		DB	75.0	80.4	88.5	98.2	104.1	106.7	107.2	106.2	103.6	98.2	86.1	76.2	(25)	
(26)		MCWB	56.5	57.0	59.8	63.2	66.3	67.3	68.5	68.9	68.0	67.3	62.5	57.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	68.3	66.9	69.1	70.9	73.1	72.4	74.7	74.1	72.9	72.9	70.8	68.1	(27)	
(28)		MCD B	82.7	89.2	95.1	103.5	107.6	107.0	106.7	107.0	105.5	99.6	91.4	83.8	(28)	
(29)		WB	63.5	63.0	65.5	68.0	70.3	70.4	71.8	72.2	71.1	70.8	67.9	63.7	(29)	
(30)		MCD B	79.8	86.3	94.4	101.2	105.3	107.1	106.7	106.3	103.5	99.6	89.3	82.0	(30)	
(31)		WB	59.6	59.8	62.8	65.7	68.5	69.3	70.4	70.9	69.9	69.1	65.5	60.9	(31)	
(32)		MCD B	77.3	83.7	89.8	98.9	103.7	105.8	105.6	105.2	102.5	98.4	87.7	78.6	(32)	
(33)		WB	56.7	57.2	60.4	64.0	66.9	68.2	69.2	69.9	68.8	67.7	63.1	58.3	(33)	
(34)		MCD B	73.9	79.2	87.3	96.5	101.9	104.2	104.2	103.9	100.7	96.3	84.5	75.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	23.2	24.6	25.8	26.6	25.8	25.7	24.2	24.1	24.9	24.3	23.6	22.6	(35)	
(36)		5% DB	27.1	30.0	29.9	30.3	28.5	27.7	25.8	25.1	26.5	26.7	26.4	25.6	(36)	
(37)		MCWBR	15.1	14.8	14.3	13.2	12.2	11.3	10.3	9.5	10.7	11.1	12.6	14.5	(37)	
(38)		5% WB	26.4	29.3	28.6	29.2	27.1	26.2	24.6	24.3	25.4	25.3	25.2	25.6	(38)	
(39)		MCWBR	16.0	15.5	14.6	13.6	12.4	11.5	10.7	10.1	11.0	11.6	13.6	15.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.370	0.407	0.449	0.534	0.552	0.486	0.506	0.525	0.494	0.473	0.393	0.371	(40)	
(41)		taud	2.200	2.062	1.936	1.747	1.726	1.899	1.882	1.849	1.906	1.949	2.150	2.197	(41)	
(42)		Ebn at Noon	279	276	270	249	244	258	253	248	253	252	268	273	(42)	
(43)		Edh at Noon	41	50	60	74	75	63	64	66	61	56	43	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1532	1815	2158	2392	2507	2585	2510	2373	2179	1871	1596	1440	(44)	
(45)		RadStd	48	52	62	57	61	31	38	41	46	28	27	25	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	+1.26	+0.81	N/A	N/A	+0.54	N/A	-103	+461	+367

Nomenclature: See separate page